

Practical No. 7

Study and implementation of node.js

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Batch - S5

Problem Statement 1: Database Connectivity using SQL / Oracle

1. Connect to Database and Retrieve Data

Answer:

- Node.js uses libraries like mysql2 or oracledb to connect to databases
- It allows executing SQL queries and retrieving results

Example:

```
const mysql = require('mysql2');
const db = mysql.createConnection({
  host: 'localhost',
  user: 'root',
  password: '',
  database: 'test'
});
db.connect((err) => {
  if (err) {
    console.log("Connection error");
    return;
  }
  db.query("SELECT * FROM users", (err, result) => {
    if (err) throw err;
    console.log(result);
  });
});
```

Key Points:

- createConnection() → establishes connection
- query() → executes SQL query
- Result is returned as array of objects

Problem Statement 2: Middleware (Express.js)

1. What is Middleware?

Answer:

- Middleware is a function that executes between request and response
- It can modify request or response objects

Uses:

- Logging
 - Authentication
 - Validation
-

2. Custom Middleware

Answer:

```
const logger = (req, res, next) => {  
  console.log(req.url);  
  next();  
};  
app.use(logger);
```

Key Points:

- req → request object
 - res → response object
 - next() → passes control to next function
-

3. Execution Order of Middleware

Answer:

- Middleware executes in the order it is defined
- Order of app.use() is important

Example:

- app.use(A)
- app.use(B)

Execution:

- A → B → Route Handler
- If next() is not called, request stops

Problem Statement 3: File System (fs Module)

1. Read and Write Files

Answer:

```
const fs = require('fs');  
fs.readFile('file.txt', 'utf8', (err, data) => {  
  console.log(data);  
});  
fs.writeFile('file.txt', 'Hello', () => {});
```

2. Difference between readFile and readFileSync

Answer:

- readFile → asynchronous, non-blocking
 - readFileSync → synchronous, blocking
-

3. Check File Exists

Answer:

```
fs.existsSync('file.txt');
```

4. Asynchronous File Handling

Answer:

- Uses callbacks, promises or async/await
 - Does not block execution
 - Improves performance
-

Problem Statement 4: File Upload and Download API

1. Upload File

Answer:

```
const multer = require('multer');
const storage = multer.diskStorage({
  destination: 'uploads/',
  filename: (req, file, cb) => {
    cb(null, Date.now() + '-' + file.originalname);
  }
});
const upload = multer({
  storage: storage,
  limits: { fileSize: 10 * 1024 * 1024 }
});
app.post('/upload', upload.single('file'), (req, res) => {
  res.send("File uploaded successfully");
});
```

2. Download File

Answer:

```
app.get('/download/:name', (req, res) => {
  const filePath = 'uploads/' + req.params.name;
  res.download(filePath);
});
```

3. Error Handling

Answer:

- Set file size limit using multer
 - Validate file type (image, pdf, etc.)
 - Handle errors using middleware or try-catch
-

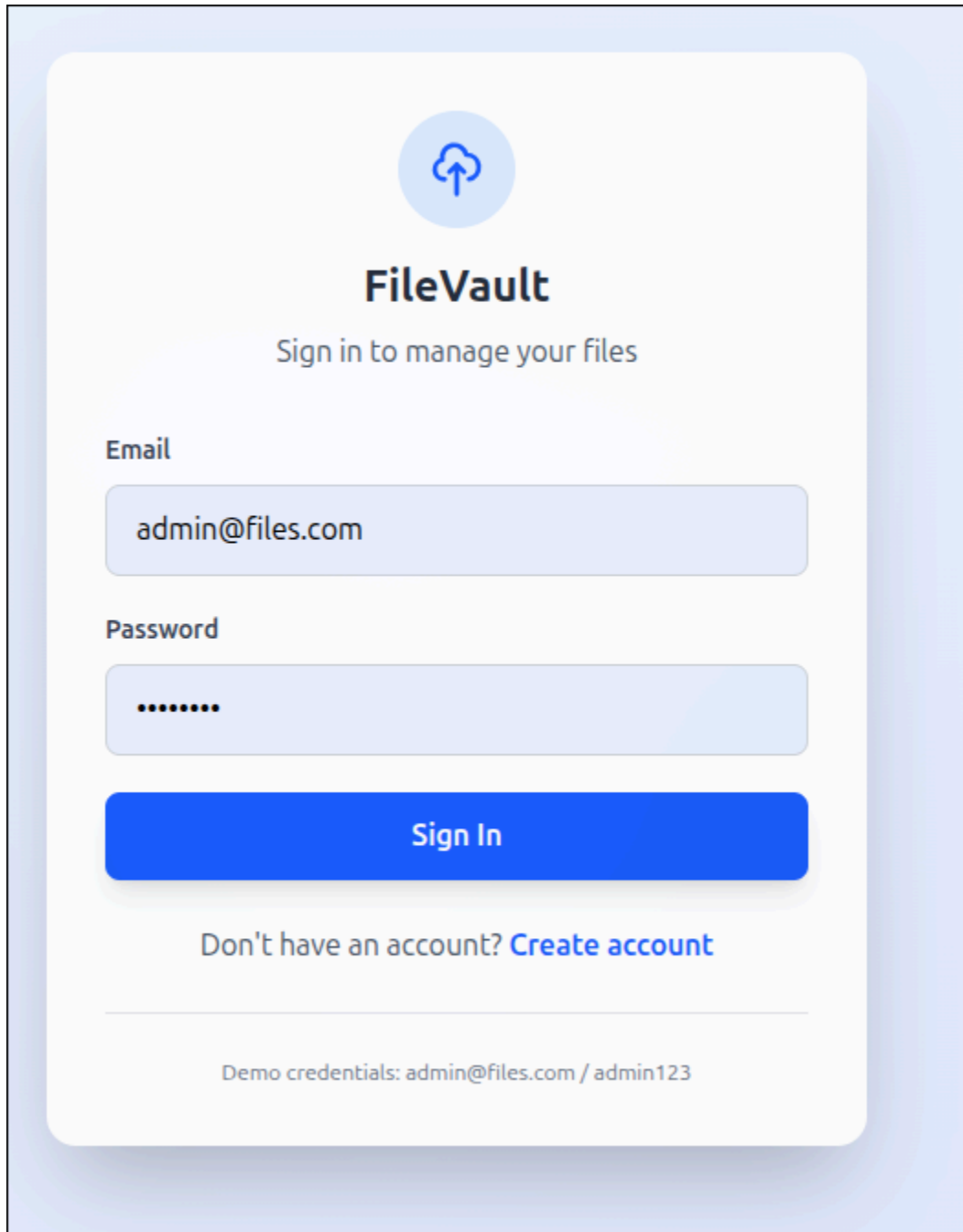
4. File Versioning

Answer:

- Prevents overwriting files with same name
- Use unique naming

Example:

- Date.now() + originalname

A login form for 'FileVault' with a light blue background and a white rounded rectangle in the center. At the top is a blue circular icon with a white cloud and an upward arrow. Below it is the text 'FileVault' in bold, followed by 'Sign in to manage your files'. There are two input fields: 'Email' with the value 'admin@files.com' and 'Password' with masked characters '.....'. A blue 'Sign In' button is below the password field. At the bottom, it says 'Don't have an account? [Create account](#)' and 'Demo credentials: admin@files.com / admin123'.

Walchand College of Engineering, Sangli
Department of Computer Science and Engineering

**FileVault**
File Management System

Admin
3 Files • 5.6 MB

Logout

Total Files
3



Storage Used
5.6 MB



Account
Admin



Upload New File



Drop files here or click to browse
Supports: JPG, PNG, GIF, PDF, DOC, DOCX (Max 10MB)

Select File

Your Files (3)

FILE	TYPE	SIZE	UPLOADED	ACTIONS
 Assignment No. 6 (1).pdf Assignment No. 6 (1)-17754...	PDF Document	184.8 KB	4/6/2026, 10:51:03 PM	Download Delete
 DBMS_Full_Notes.pdf DBMS_Full_Notes-1775496068...	PDF Document	5.4 MB	4/6/2026, 10:51:09 PM	Download Delete
 Atharv_GSoC_Proposal.pdf Atharv_GSoC_Proposal-17754...	PDF Document	2.9 KB	4/6/2026, 10:51:15 PM	Download Delete

FileVault

File Management System

Total Files

3

localhost:5175 says

Are you sure you want to delete this file? This action cannot be undone.

Cancel

OK

Admin

3 Files • 5.6 MB

Logout

Account

Admin

Upload New File

Drop files here or click to browse

Supports: JPG, PNG, GIF, PDF, DOC, DOCX (Max 10MB)

Select File

Your Files (3)

FILE	TYPE	SIZE	UPLOADED	ACTIONS
Assignment No. 6 (1).pdf Assignment No. 6 (1)-17754...	PDF Document	184.8 KB	4/6/2026, 10:51:03 PM	Download Delete
DBMS_Full_Notes.pdf DBMS_Full_Notes-1775496668...	PDF Document	5.4 MB	4/6/2026, 10:51:09 PM	Download Delete
Atharv_GSoC_Proposal.pdf Atharv_GSoC_Proposal-17754...	PDF Document	2.9 KB	4/6/2026, 10:51:15 PM	Download Delete

Problem Statement 5: Real-time Chat Application (Socket.io)

1. Setup Server

Answer:

```
const express = require('express');  
const http = require('http');  
const { Server } = require('socket.io');  
const app = express();  
const server = http.createServer(app);  
const io = new Server(server);  
server.listen(3000);
```

2. Handle User Connection

Answer:

```
io.on('connection', (socket) => {  
  console.log("User connected");  
  socket.on('disconnect', () => {  
    console.log("User disconnected");  
  });  
});
```

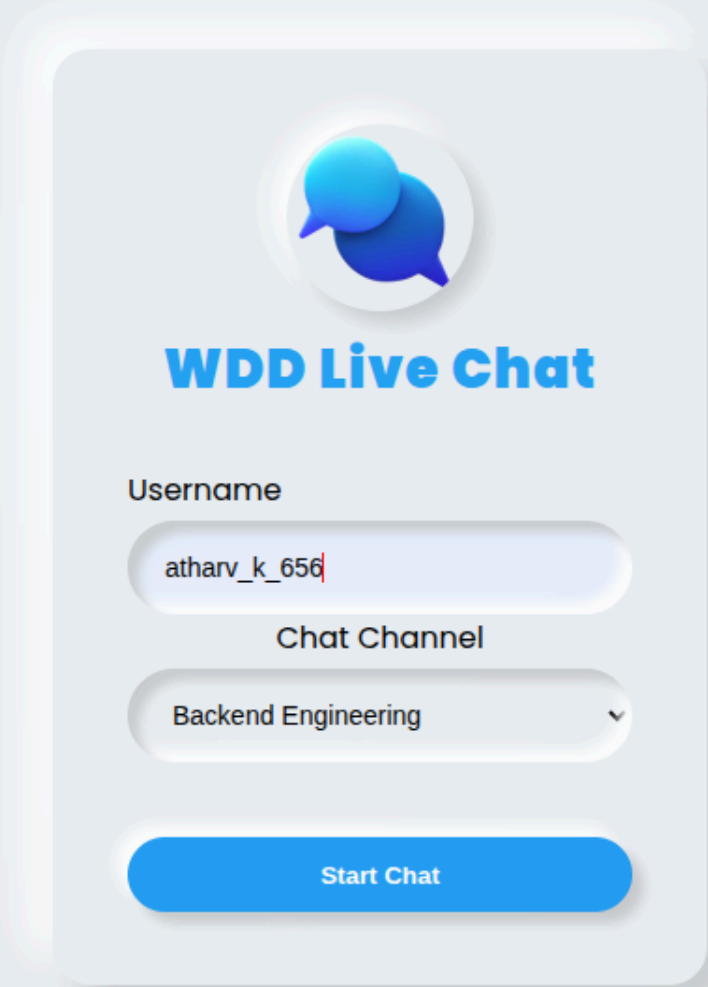
3. Message Broadcasting

Answer:

```
io.on('connection', (socket) => {  
  socket.on('message', (msg) => {  
    io.emit('message', msg);  
  });  
});
```

Key Points:

- io.on → handles connection
- socket.on → listens for events
- io.emit → sends message to all users



The image shows a web-based chat interface titled "WDD Live Chat". At the top, there is a circular icon containing two overlapping speech bubbles, one light blue and one dark blue. Below the icon, the title "WDD Live Chat" is displayed in a bold, blue, sans-serif font. Underneath the title, there are two input fields. The first is labeled "Username" and contains the text "atharv_k_656" with a red cursor at the end. The second is labeled "Chat Channel" and contains the text "Backend Engineering" with a small downward arrow on the right side, indicating a dropdown menu. At the bottom of the form is a large, rounded blue button with the text "Start Chat" in white.

